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Long-range organization of plant satellite repeats investigated using strand-specific FISH

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Abstract. The technique of chromosomal orientation and direction fluorescence in situ hybridization (COD-FISH) was adapted for plant chromosomes in order to study long-range organization of two families of satellite repeats, VicTR-B of *Vicia sativa* and PisTR-B of *Pisum sativum*. The technique allowed FISH to be performed on mitotic chromosomes in a strand-specific manner, resulting in visualization of the repeat orientation along the chromosomes and with respect to the direction of telomeric repeats. The VicTR-B probe applied to *V. sativa* chromosomes produced signals on a single chromatid at most regions containing corresponding sequences, thus confirming a presence of long arrays of head-to-tail arranged repeat monomers which is typical for satellite DNA. However, hybridization signals of different or equal intensities on both chromatids were also detected at some loci, suggesting a more complex arrangement of the repeats. Similar observations were made for PisTR-B repeats on *P. sativum* chromosomes, although the proportion of loci displaying signals on both chromatids was lower. In contrast to VicTR-B, orientation of the PisTR-B clusters with respect to telomeric sequences appeared to be conserved among subtelomeric regions of metacentric chromosomes and of the short arms of acrocentric chromosomes.

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Satellite DNA (satDNA) is composed of highly abundant genomic sequences arranged in long arrays of tandemly repeated units (monomers). SatDNA is widespread in higher plants and often makes up a significant portion of their nuclear DNA (Kato et al., 1984; Ingham et al., 1993; Irifune et al., 1995; Kubis et al., 1998; Heslop-Harrison, 2000; Macas et al., 2000). Over 160 families of the plant satellite repeats described so far are diverse in their nucleotide sequences as well as in occurrence and degree of amplification between even closely related species (Macas et al., 2002). These features reflect rapid evolutionary dynamics of this type of repetitive DNA driven by molecular mechanisms that are still poorly understood. In contrast to micro- and minisatellites, the monomers of satellite repeats are tens to thousands of nucleotides long and often form continuous arrays spanning up to 100 Mbp (Charlesworth et al., 1994; Schmidt and Heslop-Harrison, 1998). This makes their sequencing and molecular analysis over scales larger than several kilobases extremely difficult. However, such analysis is needed for investigation of molecular mechanisms causing amplification, sequence homogenization, and eventual elimination of satDNA in plant genomes. Although several theoretical models and computer simulations have been developed in order to explain these mechanisms (Smith, 1976; Walsh, 1987; Charlesworth et al., 1994; Stephan and Cho, 1994), more data about long-range composition and organization of monomers within clusters of a variety of satDNA families is needed to test their predictions or to postulate new ones.

A partial insight into the sequence composition of monomers within large satDNA clusters can be obtained using PRINS (primed in situ DNA labeling) with sequence-specific primers. In plants, this approach led to the finding that individual subfamilies of *Vicia sativa* IGS-like repeats are differentially amplified in bands present on different chromosomes (Macas et al., 2003). A mutual orientation of satellite repeat mono-

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mers can be investigated using PCR with a single primer, which can detect the presence of monomers in a head-to-head orientation (Vershinin et al., 1995), but in principle fails in assessment of their abundance and distribution. Moreover, PCR-based assays are not suitable for investigation of monomer orientation within the whole repeat cluster and of the orientation of these clusters relative to each other along the chromosomes. To achieve this goal, Goodwin and Meyne (1993) developed a technique for strand-specific hybridization of oligonucleotide probes to human mitotic chromosomes. The procedure is based on growing cells in the presence of 5-Bromo-2′-deoxyuridine (BrdU) for one replication cycle, resulting in BrdU incorporation into newly synthesized complementary strands within the chromosomes. The DNA is then made single-stranded by removing the BrdU-substituted strands. This is done by inducing DNA nicks at the sites of BrdU incorporation using UV irradiation, followed by degradation of the resulting DNA fragments with *E. coli* exonuclease III or by washing them out at elevated temperature and reduced salt concentration. This treatment yields chromosomes in which sister chromatids contain single complementary DNA strands. When hybridized with an oligonucleotide probe corresponding to a single strand of the satellite repeat, these chromosomes will display signals on a single chromatid if all monomers within the repeat cluster are in the same orientation, or on both chromatids, if the monomer orientation is mixed (Goodwin and Meyne, 1993; Meyne et al., 1994; Goodwin et al., 1996). A modification of this technique, termed chromosome orientation and direction FISH (COD-FISH), uses simultaneous hybridization of the probe of interest with a single strand of the telomeric probe which allows determination of the strand direction of the repeat under study (Meyne and Goodwin, 1995).

In this work, we adapted COD-FISH for plant chromosomes and used it to study chromosome organization and orientation of two families of satellite DNA, the VicTR-B repeats of common vetch (*Vicia sativa* L.) and the PisTR-B repeats of pea (*Pisum sativum* L.). The VicTR-B repeats contain 38-bp monomers and are amplified to millions of copies per haploid genome in several *Vicia* species, including *V. sativa* (Macas et al., 2000). In these species they form large clusters appearing as bands or dots on mitotic chromosomes (Macas et al., 2000; Navrátilová et al., 2003). Monomers within PisTR-B repeats are 50 bp long and are present in about 10,000 copies per *Pisum sativum* genome. They are localized at subtelomeric or paracentromeric regions of all pea chromosomes (Neumann et al., 2001).

Materials and methods

Plant material and chromosome preparation

Seeds from pea (*Pisum sativum* L. cv. Bohatýr) were obtained from Osiva Boršov, Czech Republic. Seeds from the common vetch (*Vicia sativa* L. cv. Ebena) were obtained from Bor s.r.o., Plzeň, Czech Republic. For preparation of chromosomes, root meristem cells of three-day-old seedlings were synchronized and accumulated in metaphase as described by Macas et al. (2000) and Gualberti et al. (1996) with the following modifications. After removal from the hydroxyurea block, the cultivation medium was supplemented with 100 μM BrdU (Boehringer Mannheim) and the cells were allowed to progress through S-phase and accumulate in metaphase in the presence of 15 μM oryzalin for 2 h *(P. sativum)* or 4 h *(V. sativa)*. The root tips were then fixed in 3% formaldehyde for 25 min at 4 °C and used for preparation of purified chromosome suspensions centrifuged onto microscopic slides (Gualberti et al., 1996). Alternatively, *V. sativa* root tips were fixed in 3:1 ethanol:acetic acid (2 h, 4 °C) and used for squash preparations as described by Leitch et al. (1994).

COD-FISH

The COD-FISH procedure generally followed the protocol described by Meyne and Goodwin (1994) and included modifications based on the results of preliminary experiments to optimize the performance of the technique with our materials. The optimization included varying proteinase K (Sigma) concentration during chromosome pretreatment (0.1–200 μg/ml), the length of UV irradiation (0–60 min), and the time of incubation with *E. coli* exonuclease III (0–45 min). The optimal conditions were as follows. The chromosome preparations were aged for 7–30 days, treated with 1–10 μg/ml proteinase K for 10 min at 37 °C as described by Leitch et al. (1994) (proteinase treatment was omitted for squash preparations), and postfixed using 4% formaldehyde for 10 min at room temperature (R.T.). The preparations were stained with 0.5 μg/ml Hoechst 33258 (Molecular Probes) in 2× SSC for 15 min at R.T. and mounted in 2× SSC. To introduce nicks into the BrdU-substituted DNA strands, the slides were exposed to 312-nm UV light (Vilber Lourmat) for 30–45 min. Then, the nicked strands were degraded by incubation with exonuclease III (3 units/μl in 50 μl of 1× reaction buffer supplied by New England Biolabs) for 5 min at 25 °C. Alternatively, the substituted strands were removed by incubation in 2× SSC for 30 min at 60 °C that caused melting of the short DNA fragments produced by UV treatment. In our experience, either of these treatments was sufficient for reproducible removal of BrdU-substituted strands (data not shown). Specificity and efficiency of the whole procedure were verified by running appropriate controls that included BrdU-free chromosomes and/or experiments with BrdU-substituted chromosomes where some of the treatments described above were omitted.

The in situ hybridizations were carried out using single-stranded oligonucleotide probes specific for telomeric (5′-(TTT AGG G)$_6$-3′), VicTR-B (5′-AAA TTT GAA GTG AAT ATA AGT CTT CAG AAA ATC TGA-3′) or PisTR-B (5′-TAA ACC CAT GAA ATT TGA TTG AAT TGT GTG AAT GAT GAA AAT-3′) repeats. The telomeric probe (purchased from Generi Biotech) was 3′ labeled with biotin, the VicTR-B and PisTR-B probes were fluorescein-labeled at their 5′ ends (Invitrogen). The hybridization mix consisted of 2× SSC, 100 ng/μl sheared calf thymus DNA, 0.125% SDS, 0.5 ng/μl telomeric probe, and 0.5 ng/μl VicTR-B or PisTR-B probe. Before hybridization, the mix was denatured at 76 °C for 15 min. Hybridization was performed overnight at 37 °C, and post-hybridization washes were performed in 2× SSC at 42 °C for 15 min. The biotin-labeled telomeric probe was detected with streptavidin-Alexa Fluor 568 (Molecular Probes). The fluorescein-labeled probes were visualized directly. Chromosomes were counterstained with DAPI (4′,6-diamidino-2-phenylindole) and examined using a Nikon Eclipse 600 microscope. The images were captured using a CCD camera and analyzed using Lucia software (Laboratory Imaging). For each tandem repeat, two independent experiments were run and at least 400 chromosomes were scored. 20–60 chromosomes of each type in *V. sativa* and 30–54 chromosomes of individual types in *P. sativum* that showed clear hybridization signals were used to construct schemes of standard patterns.

Results and discussion

The success of the COD-FISH technique largely depends on efficient and controlled incorporation of BrdU into newly synthesized strands during replication of chromosomal DNA. Thus, we chose apical root meristems of germinated seedlings as our experimental material for which efficient procedures of cell cycle synchronization had been established (Gualberti et al., 1996; Macas et al., 2000). The meristems were arrested in the G1/S phase interface using hydroxyurea and subsequently transferred to a hydroxyurea-free medium supplemented with BrdU, resulting in its incorporation into newly synthesized

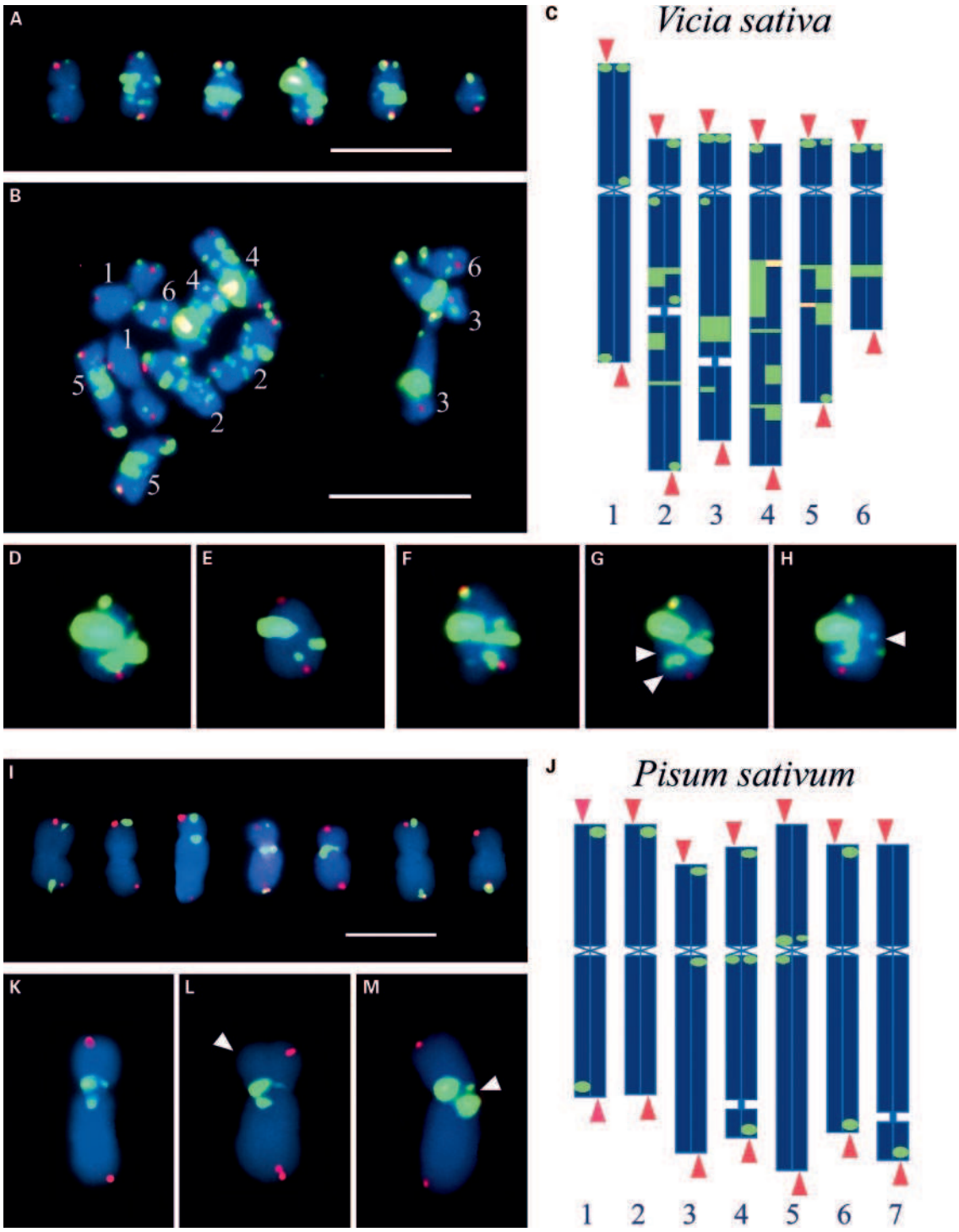


Fig. 1. COD-FISH detection of VicTR-B repeats on *V. sativa* chromosomes (**A**–**H**) and PisTR-B repeats on *P. sativum* chromosomes (**I**–**M**). The hybridization signals of VicTR-B and PisTR-B are green, the signals of the telomeric probe that specifically labels the C-rich strand (CCCTAAA)n appear red, and the chromosomes are blue due to DAPI counterstaining. Hybridization patterns typical for each chromosome are schematically represented on panels **C** *(V. sativa)* and **J** *(P. sativum)*, respectively. The green color indicates the distribution of VicTR-B and PisTR-B signals, the red arrowheads the telomeric signals at 5′ ends of chromatids. (**A**, **B**) Examples of labeling of purified chromosomes (**A**) and squash preparations (**B**) of *V. sativa* chromosomes using VicTR-B and telomeric probes. Due to considerable differences in VicTR-B abundance in individual loci (Navrátilová et al., 2003), the images were captured using lower and higher sensitivity settings (as shown for chromosome 4 in **D**, **E**) and were used for evaluating the hybridization patterns to construct the scheme given in (**C**) (yellow bands on chromosomes 4 and 5 indicate signals that appeared in less than half of scored chromosomes). Atypical labeling patterns originated by sister chromatid exchanges are shown in more detail for chromosome 4 (**G** and **H**; the supposed recombination breakpoints are marked with arrowheads) as compared to the predominant pattern (**F**). (**I**) Examples of labeling *P. sativum* chromosomes using PisTR-B and telomeric probes, and of the atypical labeling patterns observed on chromosome 5 (**L**, **M**) compared to the prevailing pattern (**K**). Bar = 10 μm.

DNA strands in the cells synchronously traversing S-phase and entering mitosis. After blocking the cells in metaphase using oryzalin, the meristems were used for chromosome preparation and optimization of subsequent treatments (UV irradiation, exonuclease III digestion or melting out nicked DNA fragments) aimed at selective removal of the BrdU-substituted strands. Purified chromosomes prepared from formaldehyde-fixed meristems were used for the experiments, as they are free of cytoplasm and cellular debris and thus easily accessible to all treatments (Doležel et al., 1992; Schubert et al., 1993). For experiments in *V. sativa*, we also used squash preparations of ethanol:acetic acid-fixed meristems (Navrátilová et al., 2003). In all experiments, the removal of BrdU-substituted DNA and strand specificity of FISH were verified by hybridization with the telomeric probe, which under optimized conditions gave signals at only one chromatid of each chromosome arm.

The COD-FISH performed on *Vicia sativa* chromosomes using the VicTR-B probe resulted in labeling all loci known to contain VicTR-B repeats (Macas et al., 2000; Navrátilová et al., 2003) but the signals clearly showed specificity to a single chromatid at most of these loci. The same labeling patterns were obtained using purified chromosomes and squash preparations (Fig. 1A, B). The localization of signals preferentially to a single chromatid was best evident for the largest VicTR-B band present on chromosome 4, as well as for a number of other interstitial loci. These observations imply that the majority of VicTR-B clusters are composed of monomers arranged in tandem (head-to-tail) orientation. However, some loci displayed signals of similar or equal intensity on both chromatids (e.g. subtelomeric loci on chromosomes 1 and 3, and intercalary loci on chromosomes 3 and 6), suggesting more complex arrangements of the repeats. Performing simultaneous hybridization of the VicTR-B sequences with the telomere-specific probe offered the advantage of determining chromosome orientation (Meyne and Goodwin, 1995). The scheme in Fig. 1C summarizes the predominant arrangement of VicTR-B signals with respect to direction of the telomeric sequences. It should be noted that other variations of these patterns could also be observed (Fig. 1G, H), most likely due to sister chromatid exchanges occurring after BrdU incorporation (Meyne and Goodwin, 1995; Cornforth and Eberle, 2001). However, the patterns shown on the scheme clearly prevailed, being present at more than 54% of scored chromosomes. As it is evident from the scheme and from Fig. 1A, B, the VicTR-B repeats are located in both orientations with respect to the telomeric signals, thus showing no preference for either direction.

In pea *(Pisum sativum)*, all chromosome loci containing PisTR-B repeats (Neumann et al., 2001) were efficiently detected using COD-FISH. Most of these loci displayed hybridization signals on only one chromatid, demonstrating that PisTR-B clusters in the pea genome are highly homogeneous with respect to orientation of the repeat monomers (Fig. 1I, J). The only exceptions were paracentromeric clusters on chromosomes 4 and 5, showing the signals on both chromatids. However, in contrast to the signals on chromosome 4 which were of the same intensity, chromosome 5 paracentromeric signals were reproducibly stronger on one chromatid (Fig. 1I, J). Examination of the signal positions with respect to signals generated by the telomeric probe revealed that the orientation of the subtelomeric PisTR-B clusters is not random but seems to be conserved among the two metacentrics (chromosomes 1 and 2) and the short arms of the three acrocentric chromosomes (3, 4, 6). At these loci, the PisTR-B probe always labeled the opposite chromatid than the probe specific for the C-rich telomere strand (Fig. 1I, J). On the other hand, the long (satellite) arms of chromosomes 4 and 7, and the long arm of chromosome 6 displayed signals of both probes on the same chromatid, thus showing that these PisTR-B clusters are in reverse orientation with respect to subtelomeric PisTR-B loci on other chromosomes (Fig. 1I, J). Similar to the observations made in *V. sativa*, the COD-FISH patterns observed in some pea chromosomes were affected by inter-chromatid recombination events (Fig. 1L, M). However, it was possible to determine the patterns typical for each chromosome type that were present on the majority of scored chromosomes (Fig. 1J).

Taken together, the data presented here demonstrate suitability of the COD-FISH technique for examination of the long-range organization of repetitive sequences on plant chromosomes. In contrast to satDNA families investigated using COD-FISH on human and mice chromosomes (Goodwin and Meyne, 1993; Meyne et al., 1994; Garagna et al., 2001), the hybridization signals obtained in our experiments were not always limited to one chromatid. Equal or similar labeling on both chromatids that was observed at some loci especially in the case of *V. sativa* VicTR-B repeats most likely indicates either (1) a mixed monomer orientation within respective repeat cluster; (2) a presence of several independent clusters composed of monomers in the same direction but arranged in mixed mutual orientations; or (3) a combination of (1) and (2). As the limited resolution of FISH on mitotic chromosomes does not allow discriminating between these possibilities, future experiments should attempt to solve this problem using prometaphase or artificially stretched chromosomes. Moreover, application of the COD-FISH technique to other families of plant satDNA and its extension to a broader range of species should show if this type of arrangement is more common and provide valuable data for comparative analysis of satDNA organization in plants.

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